Matgeo-4.8.15

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Question

A line l passes through point $(-1, 3, -2)$ and is perpendicular to both the lines $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z+1}{3}$ and $\frac{x+2}{-3} = \frac{y}{2} = \frac{z}{5}$. Find the vector equation of the line l and its distance from the origin.

Question

A line l passes through $(-1, 3, -2)$ and is perpendicular to both the lines

$$\frac{x-1}{1} = \frac{y-1}{2} = \frac{z+1}{3}, \quad \frac{x+2}{-3} = \frac{y}{2} = \frac{z}{5}. \tag{1}$$

Find the vector equation of l and its distance from the origin.

Solution

The two given direction vectors are

$$\left(d\right)_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \left(d\right)_2 = \begin{pmatrix} -3 \\ 2 \\ 5 \end{pmatrix}. \tag{2}$$

Let the required direction vector be $\left(u\right) = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$. It satisfies

$$\left(d\right)_1^\top \left(u\right) = 0, \tag{3}$$

$$\left(d\right)_2^\top \left(u\right) = 0. \tag{4}$$

Solution

Substituting values,

$$u_1 + 2u_2 + 3u_3 = 0, \tag{5}$$

$$-3u_1 + 2u_2 + 5u_3 = 0. \tag{6}$$

Eliminating u_1:

$$8u_2 + 14u_3 = 0 \quad \Rightarrow \quad u_2 = -\frac{7}{4}u_3. \tag{7}$$

Substitute in the first equation:

$$u_1 + 2\left(-\frac{7}{4}u_3\right) + 3u_3 = 0 \quad \Rightarrow \quad u_1 = \frac{1}{2}u_3. \tag{8}$$

Solution

Choosing $u_3 = 4$ gives integer values:

$$(u) = \begin{pmatrix} 2 \\ -7 \\ 4 \end{pmatrix}. \tag{9}$$

Hence the line through $(-1, 3, -2)$ is

$$(r) = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix} + t \begin{pmatrix} 2 \\ -7 \\ 4 \end{pmatrix}, \quad t \in \mathbb{R}. \tag{10}$$

Distance from Origin

For the closest point, parameter

$$t = -\frac{(P) \cdot (u)}{(u) \cdot (u)} = -\frac{(-1)(2) + 3(-7) + (-2)(4)}{2^2 + (-7)^2 + 4^2} = \frac{31}{69}. \tag{11}$$

So the foot of the perpendicular is

$$(Q) = (P) + t\,(u) = \begin{pmatrix} -\frac{7}{69} \\ \\ -\frac{10}{69} \\ \\ -\frac{14}{69} \end{pmatrix}. \tag{12}$$

Distance:

$$D = \|\,(Q)\,\| = \sqrt{\frac{5}{69}}. \tag{13}$$

Conclusion

The required line is

$$(r) = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix} + t \begin{pmatrix} 2 \\ -7 \\ 4 \end{pmatrix}, \quad t \in \mathbb{R}, \tag{14}$$

and the distance from the origin is

$$D = \sqrt{\frac{5}{69}}. \tag{15}$$

Graphical Representation